

Hospital Patient Management System

Detailed Project Structure

Bangladesh | Django REST API + React + PostgreSQL

1. Project Overview

Layer	Technology	Version	Purpose
Backend	Django + DRF	6.0.5 / 3.17.1	REST API, business logic
Database	PostgreSQL	15+	Data storage via pgAdmin4
Authentication	JWT (SimpleJWT)	5.5.1	Secure token-based auth
Frontend	React + Vite	19+ / 8+	User interface
Styling	Tailwind CSS	v4	UI design
State Mgmt	Redux Toolkit	2+	Login state management
HTTP Client	Axios	1+	API calls from React
Payments	bKash/Nagad/Rocket	SSLCommerz	Bangladesh payments

2. Project Statistics

Metric	Count
Django Apps	7
Database Tables	15+
API Endpoints	20+
React Pages	9
User Roles	5 (Admin, Doctor, Receptionist, Nurse, Lab Tech)
Payment Methods	5 (Cash, bKash, Nagad, Rocket, Card)

3. Backend Structure — server/

Technology: Django 6.0.5 + Django REST Framework 3.17.1 + PostgreSQL

Base path: D:\Lessons\ICT_Soft_Engg\hospital_pms\server\

3.1 Core Configuration — config/

config/	
settings.py	– Database, JWT, CORS, installed apps, timezone (Asia/Dhaka)
urls.py	– Root URL routing for all API endpoints
wsgi.py	– WSGI application entry point

3.2 Authentication App — authentication/

authentication/	
models.py	– Custom User model with 5 roles
serializers.py	– LoginSerializer, RegisterSerializer, UserSerializer
views.py	– LoginView, RegisterView, LogoutView, ProfileView
urls.py	– /api/auth/login/, /register/, /logout/, /profile/
admin.py	– Custom admin with role display
migrations/	– Database migration files

User Roles:

- ADMIN
- DOCTOR
- RECEPTIONIST
- NURSE
- LAB_TECHNICIAN

3.3 Patients App — patients/

patients/	
models.py	– Patient model with auto ID (P-2026-0001 format)
serializers.py	– PatientSerializer, PatientListSerializer
views.py	– CRUD operations + search + ?all=true param
urls.py	– /api/patients/ and /api/patients/<uuid:pk>/
admin.py	– Patient admin with search and filters
migrations/	– Database migration files

Key Patient Fields:

id (UUID), patient_id (auto), first_name, last_name,
date_of_birth, gender, blood_group, nid_number,
phone, emergency_contact, address, district,
allergies, chronic_conditions, is_active

3.4 Appointments App — appointments/

appointments/	
models.py	– Department, Doctor, Appointment models
serializers.py	– DepartmentSerializer, DoctorSerializer, AppointmentSerializer
views.py	– Department/Doctor/Appointment CRUD + TodayAppointmentsView
urls.py	– /api/appointments/ + /departments/ + /doctors/ + /today/
admin.py	– Admin for all 3 models
migrations/	– Database migration files

Appointment Status:

SCHEDULED, CONFIRMED, IN_PROGRESS, COMPLETED, CANCELLED, NO_SHOW

3.5 Billing App — billing/

```
billing/
  models.py      - Invoice model + InvoiceItem model
  serializers.py - InvoiceSerializer, InvoiceListSerializer,
                  InvoiceItemSerializer
  views.py       - InvoiceListCreateView, InvoiceDetailView, CollectPaymentView
  urls.py        - /api/billing/ and /api/billing/<uuid:pk>/pay/
  admin.py       - Invoice admin with inline items
  migrations/    - Database migration files
```

Payment Methods:

CASH, BKASH, NAGAD, ROCKET, CARD

Invoice Status:

PENDING, PAID, PARTIAL, CANCELLED

3.6 Prescriptions App — prescriptions/

```
prescriptions/
  models.py      - Prescription model + PrescriptionItem model
  serializers.py - PrescriptionSerializer, PrescriptionListSerializer
  views.py       - List, Create, Detail views
  urls.py        - /api/prescriptions/
  admin.py       - Prescription admin with inline medicines
  migrations/    - Database migration files
```

Medicine Frequency Options:

OD (Once daily), BD (Twice daily), TDS (3x daily),
QDS (4x daily), SOS (When needed), STAT (Immediately)

3.7 Lab App — lab/

```
lab/
  models.py      - LabTest, LabOrder, LabOrderItem models
  serializers.py - LabTestSerializer, LabOrderSerializer,
                  LabOrderListSerializer
  views.py       - LabTestListView, LabOrderListCreateView, LabOrderDetailView
  urls.py        - /api/lab/tests/ and /api/lab/orders/
  admin.py       - Lab admin with inline order items
  migrations/    - Database migration files
```

Lab Order Status:

ORDERED -> SAMPLE_COLLECTED -> IN_PROGRESS -> COMPLETED / CANCELLED

3.8 Root Backend Files

<code>manage.py</code>	– Django management commands
<code>requirements.txt</code>	– All Python dependencies
<code>.env</code>	– Environment variables (DB password, secret key)
<code>venv/</code>	– Python virtual environment (not in version control)

3.9 API Endpoints Summary

Endpoint	Methods	Description
<code>/api/auth/login/</code>	POST	Login and get JWT tokens
<code>/api/auth/logout/</code>	POST	Logout and blacklist token
<code>/api/auth/profile/</code>	GET	Get current user profile
<code>/api/auth/token/refresh/</code>	POST	Refresh access token
<code>/api/patients/</code>	GET, POST	List all / create patient
<code>/api/patients/<id>/</code>	GET, PUT, DELETE	Get / update / deactivate patient
<code>/api/appointments/</code>	GET, POST	List / book appointment
<code>/api/appointments/departments/</code>	GET, POST	Departments list
<code>/api/appointments/doctors/</code>	GET, POST	Doctors list
<code>/api/appointments/today/</code>	GET	Today's appointments
<code>/api/billing/</code>	GET, POST	List / create invoice
<code>/api/billing/<id>/pay/</code>	POST	Collect payment
<code>/api/prescriptions/</code>	GET, POST	List / create prescription
<code>/api/prescriptions/<id>/</code>	GET	Get prescription detail
<code>/api/lab/tests/</code>	GET, POST	Lab tests catalogue
<code>/api/lab/orders/</code>	GET, POST	Lab orders list / create
<code>/api/lab/orders/<id>/</code>	GET, PUT	Get / update lab order status

4. Frontend Structure — client/

Technology: React 19 + Vite 8 + Tailwind CSS v4 + Redux Toolkit

Base path: D:\Lessons\ICT_Soft_Engg\hospital_pms\client\

4.1 Root Frontend Files

<code>vite.config.js</code>	– Vite config with <code>@tailwindcss/vite</code> plugin
<code>package.json</code>	– npm dependencies
<code>index.html</code>	– HTML entry point
<code>node_modules/</code>	– Installed packages (not in version control)

4.2 Source Files — src/

src/	
App.jsx	– React Router routes + PrivateRoute guard
main.jsx	– ReactDOM.createRoot + Redux Provider + BrowserRouter
index.css	– @import 'tailwindcss' (Tailwind v4 setup)

4.3 API Layer — src/api/

src/api/	
axios.js	– Axios instance with: • baseURL: http://127.0.0.1:8000/api • JWT token auto-attach interceptor • 401 redirect to /login interceptor

4.4 State Management — src/store/

src/store/	
index.js	– Redux configureStore with all reducers
authSlice.js	– Auth state: user, token, loading, error Actions: loginUser (async), logout, clearError

4.5 Layout Components — src/components/layout/

src/components/layout/	
Layout.jsx	– Main layout wrapper with Sidebar + <Outlet />
Sidebar.jsx	– Left navigation with: • Hospital PMS logo + Bengali text • 7 NavLink menu items with active highlighting • User info (username + role) at bottom • Logout button

4.6 Pages — src/pages/

Page File	Route	Key Features
auth/LoginPage.jsx	/login	JWT login form, error display, auto-redirect
dashboard/Dashboard.jsx	/	Stats cards: patients, appointments, revenue, beds
patients/PatientList.jsx	/patients	Patient table, add form, search, auto patient ID
appointments/Appointments.jsx	/appointments	Appointment list, book form, date/status filter
billing/Billing.jsx	/billing	Invoice list, create form, payment modal, print invoice
prescriptions/Prescriptions.jsx	/prescriptions	Prescription list, write Rx form, detail modal
lab/Lab.jsx	/lab	Lab orders, test selection, status workflow
reports/Reports.jsx	/reports	Analytics: revenue, patients, departments overview

4.7 npm Dependencies

Package	Version	Purpose
react	19+	UI library
react-dom	19+	DOM rendering
react-router-dom	7+	Client-side routing
@reduxjs/toolkit	2+	State management
react-redux	9+	React-Redux binding
axios	1+	HTTP requests to Django API
tailwindcss	4+	Utility-first CSS framework
@tailwindcss/vite	4+	Tailwind v4 Vite plugin
vite	8+	Build tool and dev server

5. Database Structure — PostgreSQL

Database name: hospital_pms
Host: localhost:5432 | Tool: pgAdmin 4

5.1 Database Tables

Table Name	App	Key Fields
authentication_user	authentication	id, username, role, phone, is_active
patients_patient	patients	id, patient_id, first_name, last_name, dob, blood_group, nid
appointments_department	appointments	id, name, code, is_active
appointments_doctor	appointments	id, user_id, department_id, specialization, fee
appointments_appointment	appointments	id, patient_id, doctor_id, scheduled_at, status
billing_invoice	billing	id, invoice_number, patient_id, total, status, payment_method
billing_invoiceitem	billing	id, invoice_id, description, quantity, unit_price
prescriptions_prescription	prescriptions	id, patient_id, doctor_id, diagnosis, follow_up
prescriptions_prescriptionitem	prescriptions	id, prescription_id, medicine, dosage, frequency
lab_labtest	lab	id, name, code, price, turnaround_hours
lab_laborder	lab	id, order_number, patient_id, doctor_id, status
lab_laborderitem	lab	id, order_id, test_id, result, is_abnormal
django_migrations	django	All applied migration records
auth_group	django	Permission groups
django_session	django	User session data

6. How to Run the Project

6.1 Start Django Backend

```
cd D:\Lessons\ICT_Soft_Engg\hospital_pms\server
venv\Scripts\activate
python manage.py runserver
```

Backend running at: <http://127.0.0.1:8000>

Admin panel at: <http://127.0.0.1:8000/admin>

6.2 Start React Frontend

```
cd D:\Lessons\ICT_Soft_Engg\hospital_pms\client
npm run dev
```

Frontend running at: <http://localhost:5173>

6.3 Default Login Credentials

Field	Value
Username	admin
Password	1234 (change in production!)
Role	ADMIN
Email	admin@hospital.com

7. Completed Modules Summary

Module	Backend	Frontend	Key Feature
Authentication	JWT + Roles	Login page	5 roles, token refresh
Patients	CRUD API	List + Add form	Auto Patient ID generation
Appointments	CRUD + filter	Book + filter	Department/Doctor selection
Billing	Invoice + Pay	Invoice + Print	bKash/Nagad/Rocket support
Prescriptions	Rx + Items	Write + View	Medicine frequency, follow-up
Lab & Diagnostics	Order + Status	Order + Workflow	Sample collect to complete
Reports	Aggregated data	Analytics dashboard	Revenue, patients, departments